

Elijah Verhoff

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Education

University of Maine – B.S. Mechanical Engineering, Robotics Focus

May 2026

Experience

Lead Project Engineer (Contract), The ONIX Corporation – Remote

Jan 2026 – Jun 2026

- Prevented over \$200,000 in incorrect purchases by validating vendor quotations against 2D CAD, 3D assembly models, and P&ID documentation
- Commissioned burners, dryers, cyclones, and rotary airlocks at three customer plants in WI, OH, and Canada, performing PLC I/O mapping and HMI verification
- Cut per-unit cost of a drag-chain conveyor ~\$5,000 through design-for-manufacturability changes, replacing welded angle-iron rails with bolted joints
- Specified motors, actuators, sensors, and drives to NFPA and NEMA ratings for rotary airlocks handling explosive particulate in hazardous-dust environments
- Reviewed control schematics, electrical drawings, and PLC/HMI logic; revised 3D CAD models and BOMs to hold scope on specification

Engineering Specialist, Advanced Structures and Composites Center – Orono, ME

Dec 2024 – Dec 2025

- Led development of an in-situ winding mechanism for a continuous forming machine, sizing the rotating assembly and frame from Mathcad displacement and critical-velocity analysis
- Designed a custom slip-ring and winder assembly in SolidWorks delivering embedded power and tension-transducer feedback for closed-loop web-tension control
- Reduced material use 22% in large-area additive manufacturing by tuning overhang lengths and authoring ASTM-informed work instructions

Student Research Assistant, Advanced Manufacturing Center – Orono, ME

Jun 2018 – Sep 2018

- Produced fully dimensioned SolidWorks drawing packages with GD&T for manufacturing, inspection, and reverse engineering
- Generated CAM programs and operated 5-axis CNC mills, lathes, and laser cutters to fabricate precision parts

Projects

Rapidly Deployable Inflatable Hallway Barrier (Capstone)

elijahverhoff.github.io/projects/capstone

- Led controls and systems integration for a Sandia National Laboratories sponsored barrier that seals indoor choke points
- Achieved full deployment in 15 seconds at 50 psi through Arduino-driven pneumatic automation with pressure-feedback sensing

Neural-Network-Driven Robotic Percussionist

elijahverhoff.github.io/projects/robot-percussion-nn

- Built a robotic percussionist that strikes in time with live music, driven by a neural network trained for real-time beat tracking (0.855 test F-measure)
- Synchronized servo strikes within ~15 ms, below the human just-noticeable threshold, via an end-to-end audio-to-Arduino pipeline

Skills

Mechanical Design: SolidWorks, Rhinoceros 3D, Fusion 360, GD&T, design for manufacturability

Engineering Analysis: Abaqus, MATLAB, Python, PyTorch

Robotics and Controls: ROS2, robot kinematics, motion planning, vision systems, PLC/HMI logic, closed-loop control

Manufacturing: CNC machining, CAM programming, prototyping, additive manufacturing